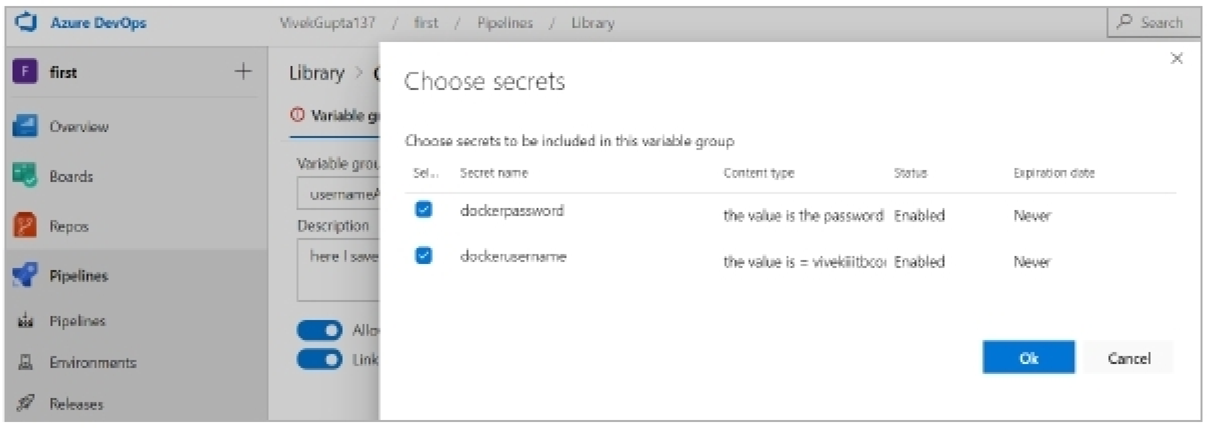

Figure 12: Both secrets from key vault to variable group have been selected

```
1 pool:
2   name: Hosted Ubuntu 1604
3
4 steps:
5 - task: AzureKeyVault@1
6   inputs:
7     azureSubscription: 'Azure for Students (301fcd09-edf0-427e-86e3-11fbfaa9b9cf)'
8     KeyVaultName: 'vivekiiitbkeyvault'
9     SecretsFilter: '*'
10    RunAsPreJob: true
11
12 - script: |
13     docker build -t osfy-proj:latest .
14     docker tag osfy-proj:latest vivekiiitbcontainerregistry.azurecr.io/$(Build.Repository.Name):latest
15     docker image ls -a
16     docker login -u $(dockerusername) -p $(dockerpassword) vivekiiitbcontainerregistry.azurecr.io
17     docker push vivekiiitbcontainerregistry.azurecr.io/$(Build.Repository.Name):latest
18   displayName: 'Docker build and push'
```

Figure 13: Final *azure-pipelines.yml* file

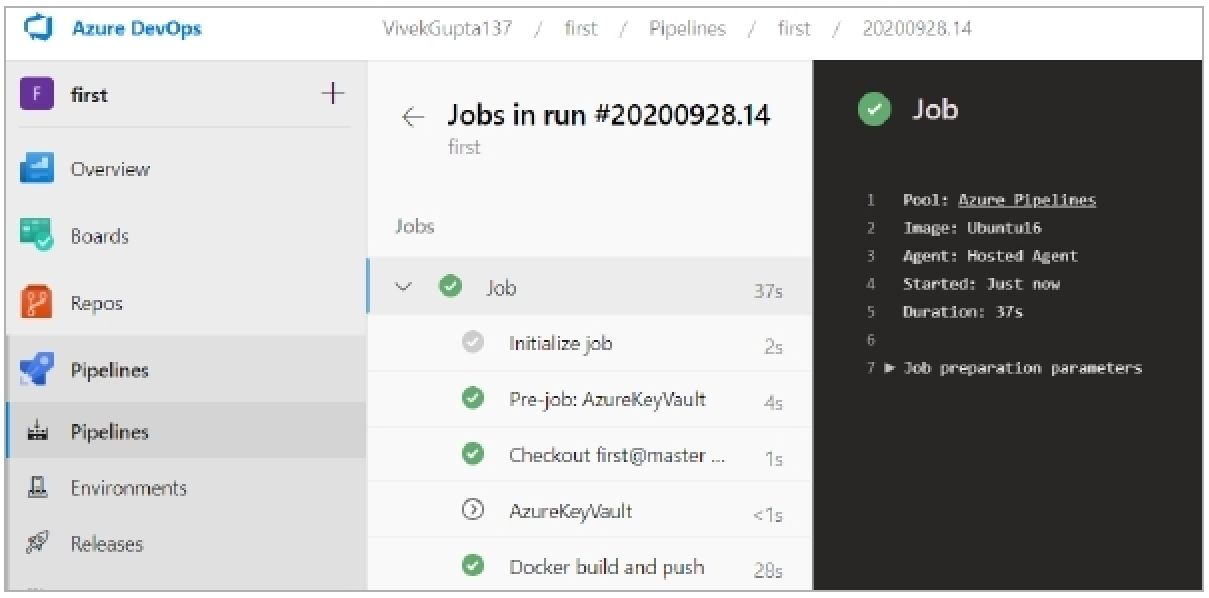

Figure 14: Logs generated after running the pipeline

configuration shown below:

```
$ az keyvault create --location eastus
--name vivekiiitbkeyvault --resource-
group theResourceGroup --sku standard
```

Updating the pipeline code

1. First, visit the Azure key vault at *https://portal.azure.com/* and navigate to the *Secrets* tab; then hit *Generate* to create a new secret and two secrets for the Azure container registry's user name (*dockerusername*) and password (*dockerpassword*), as shown in Figure 9.
2. Once done, you'll see that two secrets have been created (Figure 10).
3. Now, visit the library section of pipelines and click on *Add variable group* to create a group with the name *usernameAndpassword*, as shown in Figure 11.
4. Click on *Add variables* to get the pop-up to choose the secrets and then click the *OK* button, as shown in Figure 12.
5. Now navigate back to the pipelines section, and edit the code of *azure-pipelines.yaml* to remove the user name and passwords using variables, as shown in Figure 13.

The Docker image has been built using this YAML file. Later, push it into the Azure container registry. Lines 5-10 are added, and line 16 is updated.

- *Lines 5-10:* These lines add a new task, which adds the Azure key vault to the pipeline. This task will run before the scripts. Here, you have specified the Azure subscription to use the name of the key vault that was created earlier -- *SecretsFilter with value* = '*'. It will choose all the secrets in the key vault.
- *Line 16:* Here the actual user name and password are replaced with the pipeline variables defined in the *usernameAndPassword* variable group.

6. Now run this pipeline by clicking the *Run* button. Once the job runs completely, you can see the logs, as seen in Figure 14.

We have now successfully removed the user name and passwords from our pipeline.

For faster software builds and delivery, CI/CD can be used to automate builds and deployments.

We do hope this article has helped you understand Azure DevOps and the procedure to build a CI/CD pipeline. END

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